

1. Introduction

"Since the launch of ChatGPT in November 2022, the landscape of AI technologies has undergone a significant transformation. These technologies have reshaped interactions and have become deeply integrated into various aspects of daily life and industry.

In February 2024, building on this momentum, OpenAI introduced Sora, a text-to-video generative AI model. Sora stands out from previous video generation models due to its ability to generate videos of up to one minute in length, with high quality, and while strictly adhering to the user's text instructions.

The development of Sora embodies the long-standing mission of AI research: to equip AI systems, or AI Agents, with the ability to understand and interact with the physical world in motion. This involves the development of AI models that can interpret complex user instructions and apply this understanding to solve real-world problems through dynamic and contextually rich simulations.

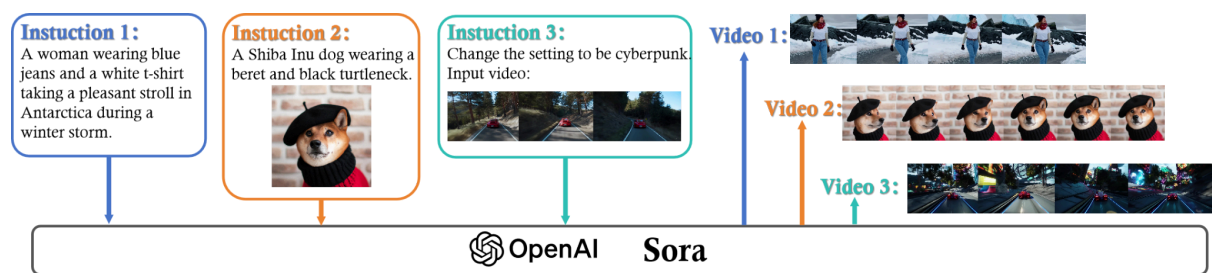


Figure 2: Examples of Sora in text-to-video generation. Text instructions are given to the OpenAI Sora model, and it generates three videos according to the instructions.

Sora demonstrates a remarkable ability to accurately interpret and execute complex human instructions, as illustrated in Figure 2. The model can generate detailed scenes that include multiple characters performing specific actions against intricate backgrounds.

The proficiency of Sora is attributed not only to its ability to process user-generated textual prompts but also to its capacity to discern the complicated interplay of elements within a scenario. One of the most striking aspects of Sora is its ability to generate up to a minute-long video while maintaining high visual quality and compelling visual coherency.

Unlike earlier models that could only generate short video clips, Sora's minute-long video creation offers a sense of progression and a visually consistent journey from the first frame to the last. Furthermore, Sora's advancements are evident in its ability to produce extended video sequences with nuanced depictions of motion and interaction. This overcomes the constraints of shorter clips and simpler visual renderings that characterized earlier video generation models.

This capability represents a significant leap forward in AI-driven creative tools, enabling users to convert text narratives into rich visual stories. Overall, these advances highlight the potential of Sora as a world simulator, providing nuanced insights into the physical and contextual dynamics of the depicted scenes."

2. History

Before the rise of deep learning, creating images with computers was a bit like drawing by hand. We used methods like texture synthesis and texture mapping, but these methods had their limits and couldn't create very complex or vivid images.

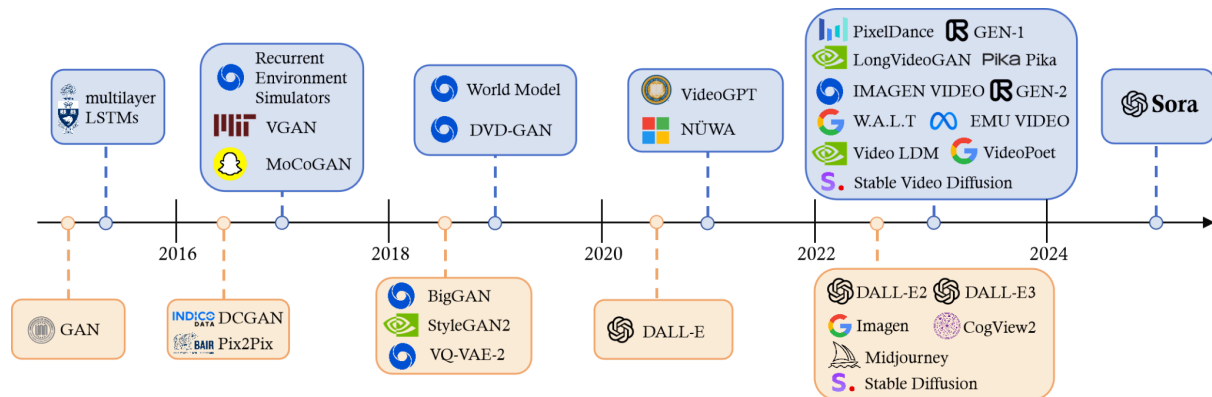


Figure 3: History of Generative AI in Vision Domain.

Then came along some game-changers: Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs). These tools were a big step forward and allowed us to create images with more detail and quality. This progress has made it easier for anyone to create content using simple text instructions.

Over the past ten years, we've seen a lot of different approaches to creating images with computers. One of the most successful has been the transformer architecture, which was first used in language processing. Researchers have now started using it in computer vision tasks too, leading to models like Vision Transformer (ViT) and Swin Transformer.

At the same time, diffusion models have also improved image and video generation. These models use a mathematical framework to turn noise into images, with U-Nets helping to predict and reduce noise at each step.

Since 2021, there's been a big focus on models that can understand human instructions. These models, known as multimodal models, can work with both text and images. For example, CLIP is a model that combines transformer architecture with visual elements, allowing it to be trained on large datasets of text and images. Another example is Stable Diffusion, a versatile text-to-image AI model that can produce images in a wide range of styles.

Following the release of ChatGPT in November 2022, we have witnessed the emergence of commercial text-to-image products in 2023, such as Stable Diffusion, Midjourney, DALL-E 3. These tools enable users to generate new images of high resolution and quality with simple text prompts, showcasing the potential of AI in creative image generation. However, transitioning from text-to-image to text-to-video is challenging due to the temporal complexity of videos. Despite numerous efforts in industry and academia, most existing video generation tools, such as Pika and Gen-2, are limited to producing only short video clips of a few seconds. In this context, Sora represents a significant breakthrough, akin to

ChatGPT's impact in the NLP domain. Sora is the first model that is capable of generating videos up to one minute long based on human instructions, marking a milestone that profoundly influences research and development in generative AI.

3. Applications

The implications of this technology extend far beyond mere video creation, offering transformative potential for tasks ranging from automated content generation to complex decision-making processes. In this section, we delve into a comprehensive examination of the current applications of video diffusion models, highlighting key areas where Sora has not only demonstrated its capabilities but also revolutionized the approach to solving complex problems.

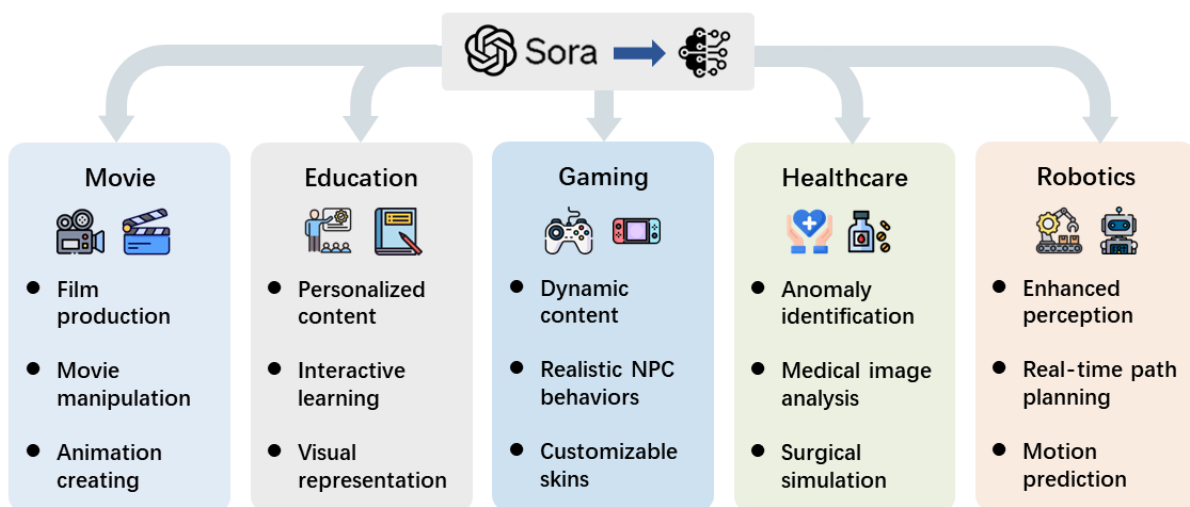


Figure 4: Applications of Sora.

We aim to offer a broad perspective for the practical deployment scenarios (see Figure 4).

3.1 Movies

Traditionally, creating cinematic masterpieces has been an arduous and expensive process, often requiring decades of effort, cutting-edge equipment, and substantial financial investments. However, the advent of advanced video generation technologies heralds a new era in film-making, one where the dream of autonomously producing movies from simple text inputs is becoming a reality. Researchers have ventured into the realm of movie generation by extending video generation models into creating movies. MovieFactory applies diffusion models to generate film-style videos from elaborate scripts produced by Chat GPT, representing a significant leap forward. In the follow-up, MobileVidFactory can automatically generate vertical mobile videos with only simple texts provided by users. Vlogger makes it feasible for users to compose a minute-long vlog. These developments, epitomized by Sora's ability to generate captivating movie content effortlessly, mark a pivotal moment in the democratization of movie production. They offer a glimpse into a future where anyone can be a filmmaker, significantly lowering the barriers to entry in the film industry and introducing a novel dimension to movie production that blends traditional storytelling with AI-driven creativity. The implications of these technologies extend beyond simplification.

They promise to reshape the landscape of film production, making it more accessible and versatile in the face of evolving viewer preferences and distribution channels.

3.2 Education

The landscape of educational content has long been dominated by static resources, which, despite their value, often fall short of catering to the diverse needs and learning styles of today's students. Video diffusion models stand at the forefront of an educational revolution, offering unprecedented opportunities to customize and animate educational materials in ways that significantly enhance learner engagement and understanding. These advanced technologies enable educators to transform text descriptions or curriculum outlines into dynamic, engaging video content tailored to the specific style, and interests of individual learners. Moreover, image-to-video editing techniques present innovative avenues for converting static educational assets into interactive videos, thereby supporting a range of learning preferences and potentially increasing student engagement. By integrating these models into educational content creation, educators can produce videos on a myriad of subjects, making complex concepts more accessible and captivating for students. The use of Sora in revolutionizing the educational domain exemplifies the transformative potential of these technologies. This shift towards personalized, dynamic educational content heralds a new era in education.

3.3 Gaming

The gaming industry constantly seeks ways to push the boundaries of realism and immersion, yet traditional game development often grapples with the limitations of pre-rendered environments and scripted events. The generation of dynamic, high-fidelity video content and realistic sound by diffusion models effects in real-time, promise to overcome existing constraints, offering developers the tools to create evolving game environments that respond organically to player actions and game events. This could include generating changing weather conditions, transforming landscapes, or even creating entirely new settings on the fly, making game worlds more immersive and responsive. Some methods also synthesize realistic impact sounds from video inputs, enhancing game audio experiences. With the integration of Sora within the gaming domain, unparalleled immersive experiences that captivate and engage players can be created. How games are developed, played, and experienced will be innovated, as well as opening new possibilities for storytelling, interaction, and immersion.

3.4 Healthcare

Despite generative capabilities, video diffusion models excel in understanding and generating complex video sequences, making them particularly suited for identifying dynamic anomalies within the body, such as early cellular apoptosis, skin lesion progression, and irregular human movements, which are crucial for early disease detection and intervention strategies. Additionally, models like MedSegDiff-V2 [161] and [162] leverage the power of transformers to segment medical images with unprecedented precision, enabling clinicians to pinpoint areas of interest across various imaging modalities with enhanced accuracy. The integration of Sora into clinical practice promises not only to refine diagnostic processes but also to personalize patient care, offering tailored treatment plans based on precise medical imaging analysis.

However, this technological integration comes with its own set of challenges, including the need for robust data privacy measures and addressing ethical considerations in healthcare.

3.5 Robotics

Video diffusion models now play important roles in robotics, showing a new era where robots can generate and interpret complex video sequences for enhanced perception [163, 164] and decision-making [165, 166, 167]. These models unlock new capabilities in robots, enabling them to interact with their environment and execute tasks with unprecedented complexity and precision. The introduction of web-scale diffusion models to robotics [163] showcases the potential for leveraging large-scale models to enhance robotic vision and understanding. Latent diffusion models are employed for language-instructed video prediction [168], allowing robots to understand and execute tasks by predicting the outcome of actions in video format. Furthermore, the reliance on simulated environments for robotics research has been innovatively addressed by video diffusion models capable of creating highly realistic video sequences [169, 170]. This enables the generation of diverse training scenarios for robots, mitigating the limitations imposed by the scarcity of real-world data. We believe, the integration of technologies like Sora into the robotics field holds the promise of groundbreaking developments. By harnessing the power of Sora, the future of robotics is poised for unprecedented advancements, where robots can seamlessly navigate and interact with their environments.

4. Advantages

1. **Data Handling:** Sora AI can handle data in its native size, eliminating the need to resize, crop, or trim videos to standardized dimensions. This leads to better framing and composition, and enhanced language understanding.
2. **Text-to-Video Transformation:** Sora AI can transform text instructions into realistic videos. This feature can be particularly useful in fields like education and entertainment.
3. **Detailed Scene Generation:** Sora AI can generate detailed scenes involving multiple characters and movements, providing a richer storytelling canvas for content creators.
4. **Efficiency:** Sora AI's ability to generate videos from text can significantly reduce the time and resources required for video production, making it a cost-effective solution.
5. **Innovation:** As a cutting-edge technology, Sora AI is pushing the boundaries of what's possible in artificial intelligence and machine learning. It's a step towards more interactive and immersive digital experiences.
6. **Scalability:** Sora AI's approach to handling data in its native size allows it to scale effectively. As the volume of data increases, Sora AI can continue to process and generate videos without significant degradation in performance or quality.
7. **Customizability:** With its ability to transform text instructions into videos, Sora AI offers a high degree of customizability. Users can specify their requirements in the text instructions, and Sora AI can generate videos that closely match these specifications.
8. **Interactivity:** Sora AI's text-to-video transformation capability opens up possibilities for interactive content creation. Users can modify their text instructions based on feedback or new ideas, and Sora AI can generate updated videos accordingly. This interactivity can lead to a more dynamic and engaging content creation process.
9. **Video Editing:** Sora AI can edit already made videos. This means you can input an existing video and provide text instructions to modify it. This could include adding new elements, changing the sequence of scenes, or altering the actions of characters. This flexibility makes Sora AI a powerful tool for video editing.
10. **Video Length Extension:** Sora AI has the capability to extend the length of videos. By providing additional text instructions, you can create new scenes or extend existing ones, effectively increasing the duration of the video. This can be done for both past and future scenes, giving you control over the narrative timeline.
11. **No Coding Knowledge Required:** One of the biggest advantages of Sora AI is its user-friendly interface. You don't need any coding knowledge to use Sora AI. The process of creating or editing videos involves providing text instructions, which is as simple as writing a description or a story. This makes Sora AI accessible to a wide range of users, not just those with technical or programming skills.
12. **Versatile Input Handling:** Sora AI can be prompted with a variety of inputs, including pre-existing images or videos. This capability enables Sora to perform a wide range of image and video editing tasks. For instance, it can create perfectly looping videos, animate static images, and extend videos forwards or backwards in time. This versatility in handling different types of inputs and performing diverse editing tasks makes Sora AI a comprehensive tool for multimedia content creation and manipulation.
13. **Image generation capabilities:** Sora is also capable of generating images. We do this by arranging patches of Gaussian noise in a spatial grid with a temporal extent of one frame. The model can generate images of variable sizes—up to 2048x2048 resolution.

5. Disadvantages

1. **Challenges in Physical Realism:** Sora, as a simulation platform, exhibits a range of limitations that undermine its effectiveness in accurately depicting complex scenarios. Most important is its inconsistent handling of physical principles within complex scenes, leading to a failure in accurately copying specific examples of cause and effect. For instance, consuming a portion of a cookie might not result in a corresponding bite mark, illustrating the system's occasional departure from physical plausibility. This issue extends to the simulation of motion, where Sora generates movements that challenge realistic physical modeling, such as unnatural transformations of objects or the incorrect simulation of rigid structures like chairs, leading to unrealistic physical interactions. The challenge further increases when simulating complex interactions among objects and characters, occasionally producing outcomes that lean towards the humorous.
2. **Spatial and Temporal Complexities:** Sora occasionally misunderstands instructions related to the placement or arrangement of objects and characters within a given prompt, leading to confusion about directions (e.g., confusing left for right). Additionally, it faces challenges in maintaining the temporal accuracy of events, particularly when it comes to adhering to designated camera movements or sequences. This can result in deviating from the intended temporal flow of scenes. In complex scenarios that involve a multitude of characters or elements, Sora has a tendency to insert irrelevant animals or people. Such additions can significantly change the originally envisioned composition and atmosphere of the scene, moving away from the planned narrative or visual layout. This issue not only affects the model's ability to accurately recreate specific scenes or narratives but also impacts its reliability in generating content that closely aligns with the user's expectations and the coherence of the generated output.
3. **Limitations in Human-computer Interaction (HCI):** Sora, while showing potential in the video generation domain, faces significant limitations in HCI. These limitations are primarily evident in the coherence and efficiency of user-system interactions, especially when making detailed modifications or optimizations to generated content. For instance, users might find it difficult to precisely specify or adjust the presentation of specific elements within a video, such as action details and scene transitions. Additionally, Sora's limitations in understanding complex language instructions or capturing subtle semantic differences could result in video content that does not fully meet user expectations or needs. These shortcomings restrict Sora's potential in video editing and enhancement, also impacting the overall satisfaction of the user experience.
4. **Usage Limitation:** Regarding usage limitations, OpenAI has not yet set a specific release date for public access to Sora, emphasizing a cautious approach towards safety and readiness before broad deployment. This indicates that further improvements and testing in areas such as security, privacy protection, and content review may still be necessary for Sora. Moreover, at present, Sora can only generate

videos up to one minute in length, and according to published cases, most generated videos are only a few dozen seconds long. This limitation restricts its use in applications requiring longer content display, such as detailed instructional videos or in-depth storytelling. This limitation reduces Sora's flexibility in the content creation.

5. **Misinformation and Trust Erosion:** There's a real risk of Sora producing content that could mislead or harm. The potential for creating ultra-realistic videos raises alarms about misinformation and the erosion of trust in digital content.
6. **Factual Inaccuracies and Biased Representations:** Issues like factual inaccuracies, privacy breaches, and biased representations are significant concerns. These issues can undermine the effectiveness of the generated content and potentially lead to misuse.
7. **Regulatory and Ethical Concerns:** Before Sora is released to the public, there needs to be stricter regulations and guardrails when it comes to technology. This includes considerations around security, privacy protection, and content review.
8. **Potential for Misuse:** With the advent of deepfakes and the increasing sophistication of AI-generated imagery, there's a growing concern about the potential misuse of technologies like Sora. This includes the creation of misleading or harmful content, which can have serious implications in areas like politics, personal privacy, and social trust.

6. Technology Behind Sora

6.1 Overview of Sora

In the core essence, Sora is a diffusion transformer with flexible sampling dimensions as shown in Figure 5. It has three parts:

- 1) A time-space compressor first maps the original video into latent space.
- 2) A ViT then processes the tokenized latent representation and outputs the denoised latent representation.
- 3) A CLIP-like [26] conditioning mechanism receives LLM-augmented user instructions and potentially visual prompts to guide the diffusion model to generate styled or

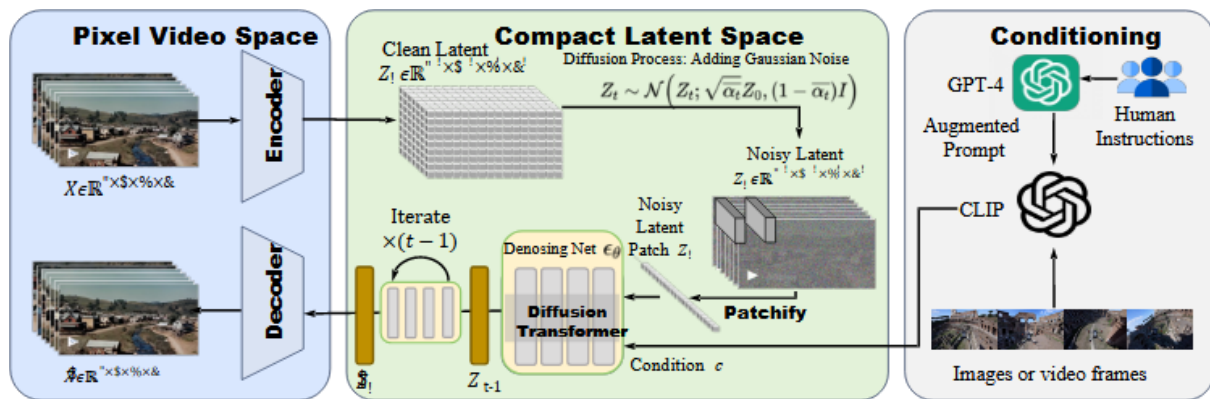


Figure 5: Overview of Sora framework

themed videos. After many denoising steps, the latent representation of the generated video is obtained and then mapped back to pixel space with the corresponding decoder. In this section, we aim to reverse engineer the technology used by Sora and discuss a wide range of related works.

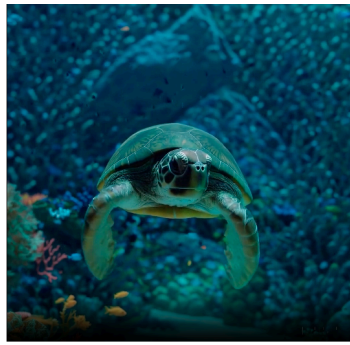
6.2 Data Pre-processing

6.2.1 Variable Durations, Resolutions, Aspect Ratios

One distinguishing feature of Sora is its ability to train on, understand, and generate videos and images at their native sizes as illustrated in Figure 6. Traditional methods often resize, crop, or adjust the aspect ratios of videos to fit a uniform standard—typically short clips with square frames at fixed low resolutions. Those samples are often generated at a wider temporal stride and rely on separately trained frame-insertion and resolution-rendering models as the final step, creating inconsistency across the video. Utilizing the diffusion transformer architecture (see Section 6.2.4), Sora is the first model to embrace the diversity of visual data and can sample in a wide array of video and image formats, ranging from widescreen 1920x1080p videos to vertical 1080x1920p videos and everything in between without compromising their original dimensions.



(a) Vertical



(b) Square



(c) Horizontal

Figure 6: Sora can generate images in flexible sizes or resolutions ranging from 1920x1080p to 1080x1920p and anything in between.

Training on data in their native sizes significantly improves composition and framing in the generated videos. Empirical findings suggest that by maintaining the original aspect ratios, Sora achieves a more natural and coherent visual narrative. The comparison between Sora and a model trained on uniformly cropped square videos demonstrates a clear advantage as shown in Figure 7. Videos produced by Sora exhibit better framing, ensuring subjects are fully captured in the scene, as opposed to the sometimes truncated views resulting from square cropping. This nuanced



(a) Training on videos that are cropped to squares leads to unnatural compositions and framing.



(b) Training in native sizes improves framing.

Figure 7: A comparison between Sora (right) and a modified version of the model (left), which crops videos to square shapes—a common practice in model training—highlights the advantages.

understanding and preservation of original video and image characteristics mark a significant advancement in the field of generative models. Sora’s approach not only showcases the potential for more authentic and engaging video generation but also highlights the importance of diversity in training data for achieving high-quality results in generative AI. The training approach of Sora aligns with the core tenet of Richard Sutton’s THE BITTER LESSON, which states that leveraging computation over human-designed features leads to more effective and flexible AI systems. Just as the original design of diffusion transformers seeks simplicity and scalability, Sora’s strategy of training on data at their native sizes eschews traditional AI reliance on human-derived abstractions, favoring instead a generalist method that scales with computational power.

6.2.2 Unified Visual Representation

To effectively process diverse visual inputs including images and videos with varying durations, resolutions, and aspect ratios, a crucial approach involves transforming all forms of visual data into a unified representation, which facilitates the large-scale training of generative models. Specifically, Sora patchifies videos by initially compressing videos into a

lower-dimensional latent space, followed by decomposing the representation into spacetime patches.

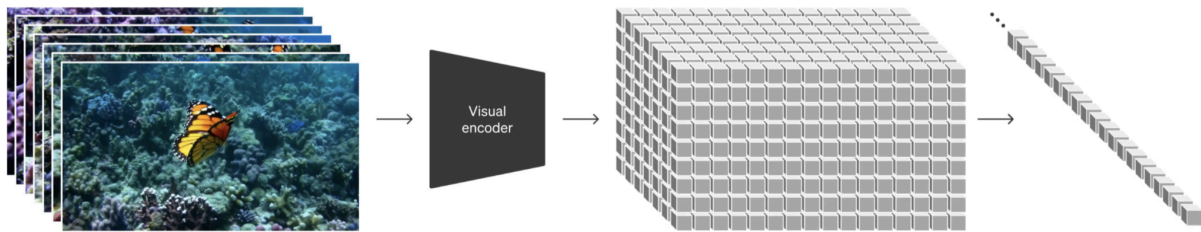


Figure 8: At a high level, Sora turns videos into patches by first compressing videos into a lower dimensional latent space, and subsequently decomposing the representation into spacetime patches.

6.2.3 Video Compression Network

Sora's video compression network (or visual encoder) aims to reduce the dimensionality of input data, especially a raw video, and output a latent representation that is compressed both temporally and spatially as shown in Figure 8.

Spatial-patch Compression: This involves transforming video frames into fixed-size patches, akin to the methodologies employed in ViT (Vision Transformer) and MAE (Masked autoencoder) (see Figure 9), before encoding them into a latent space. This approach is particularly effective for accommodating videos of varying resolutions and aspect ratios, as it encodes entire frames through the processing of individual patches. Subsequently, these spatial tokens are organized in a temporal sequence to create a spatial-temporal latent representation. This technique highlights several critical considerations:



Figure 9: ViT splits an image into fixed size patches, linearly embeds each of them, adds position embeddings, and feeds the resulting sequence of vectors to a standard Transformer encoder.

Temporal dimension variability– given the varying durations of training videos, the temporal dimension of the latent space representation cannot be fixed. To address this, one can either sample a specific number of frames (padding or temporal interpolation may be needed for much shorter videos) or define a universally extended (super long) input length for subsequent processing (more details are described in Section 6.2.4); Utilization of pre-trained visual encoders– for processing videos of high resolution, leveraging existing pre-trained visual encoders, such as the VAE encoder from Stable Diffusion, is advisable for most researchers while Sora's team is expected to train their own compression network with a decoder (the video generator) from scratch via the manner employed in training latent diffusion models. These encoders can efficiently compress large-size patches (e.g., 256×256), facilitating the management of large-scale data; Temporal information aggregation– since this method primarily focuses on spatial patch compression, it necessitates an additional mechanism for aggregating temporal information within the model. This aspect is

crucial for capturing dynamic changes over time and is further elaborated in subsequent sections (see details in Section 6.3.1 and Figure 14).

Spatial-temporal-patch Compression: This technique is designed to encapsulate both spatial and temporal dimensions of video data, offering a comprehensive representation. This technique extends beyond merely analyzing static frames by considering the movement and changes across frames, thereby capturing the video's dynamic aspects. The utilization of 3D convolution emerges as a straightforward and potent method for achieving this integration. The graphical illustration and the comparison against pure spatial-patchifying are depicted in Figure 10. Similar to spatial-patch compression, employing spatial temporal-patch compression with predetermined convolution kernel parameters— such as fixed kernel sizes, strides, and output channels— results in variations in the dimensions of the latent space due to the differing characteristics of video inputs. This variability is primarily driven by the diverse durations and resolutions of the videos being processed. To mitigate this challenge, the approaches adopted for spatial patchification are equally applicable and effective in this context.

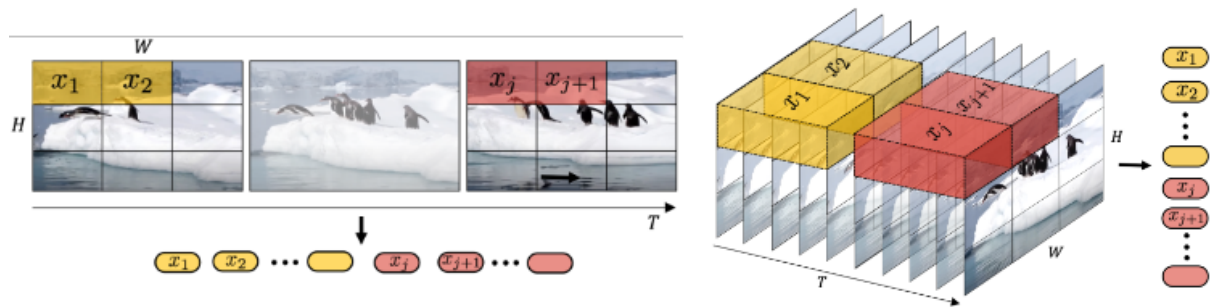


Figure 10: Comparison between different patchification for video compression. Source: ViViT. (Left) Spatial patchification simply samples nt frames and embeds each 2D frame independently following ViT. (Right) Spatial-temporal patchification extracts and linearly embeds non-overlapping or overlapping tubelets that span the spatiotemporal input volume.

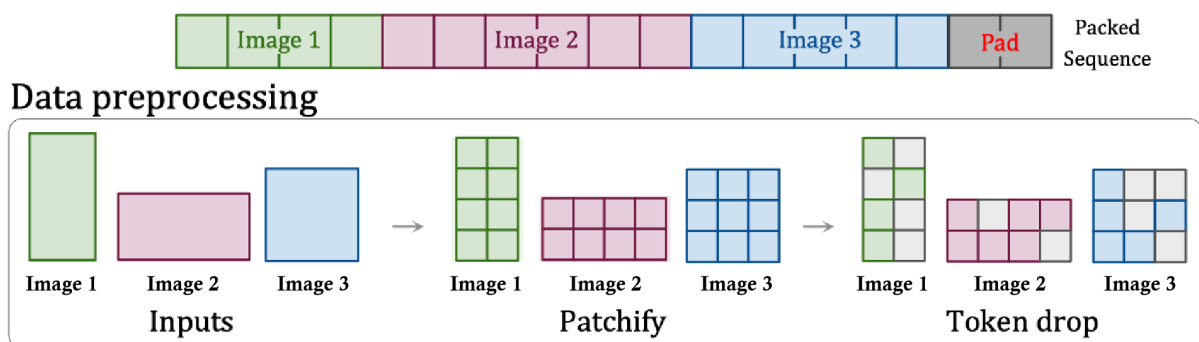


Figure 11: Patch packing enables variable resolution images or videos with preserved aspect ratio. Token dropping somehow could be treated as data augmentation.

6.2.4 Spacetime Latent Patches

There is a pivotal concern remaining in the compression network part: How to handle the variability in latent space dimensions (i.e., the number of latent feature chunks or patches

from different video types) before feeding patches into the input layers of the diffusion transformer.

Patch n' Pack (PNP) packs multiple patches from different images in a single sequence as shown in Figure 11. This method is inspired by example packing used in natural language processing that accommodates efficient training on variable length inputs by dropping tokens. Here the patchification and token embedding steps need to be completed in the compression network, but Sora may further patchify the latent for transformer token as Diffusion Transformer does. Regardless of whether there is a second-round patchification or not, we need to address two concerns, how to pack those tokens in a compact manner and how to control which tokens should be dropped. For the first concern, a simple greedy approach is used which adds examples to the first sequence with enough remaining space. Once no more examples can fit, sequences are filled with padding tokens, yielding the fixed sequence lengths needed for batched operations. Such a simple packing algorithm can lead to significant padding, depending on the distribution of the length of inputs. On the other hand, we can control the resolutions and frames we sample to ensure efficient packing by tuning the sequence length and limiting padding. For the second concern, an intuitive approach is to drop the similar tokens or, like PNP, apply dropping rate schedulers. However, it is worth noting that **3D Consistency** is one of the nice properties of Sora. Dropping tokens may ignore fine-grained details during training. Thus, we believe that OpenAI is likely to use a super long context window and pack all tokens from videos although doing so is computationally expensive e.g., the multi-head attention operator exhibits quadratic cost in sequence length. Specifically, spacetime latent patches from a long-duration video can be packed in one sequence while the ones from several short-duration videos are concatenated in the other sequence.

6.3 Modeling

6.3.1 Diffusion Transformer

Image Diffusion Transformer: Traditional diffusion models mainly leverage convolutional U-Nets that include downsampling and upsampling blocks for the denoising network backbone. However, recent studies show that the U-Net architecture is not crucial to the good performance of the diffusion model. By incorporating a more flexible transformer architecture, the transformer-based diffusion models can use more training data and larger model parameters. Along this line, DiT and U-ViT are among the first works to employ vision transformers for latent diffusion models. As in ViT, DiT employs a multi-head self attention layer and a pointwise feed-forward network interlaced with some layer norm and scaling layers. Moreover, as shown in Figure 12, DiT incorporates conditioning via adaptive layer norm (AdaLN) with an additional MLP layer for zero-initializing, which initializes each residual block as an identity function and thus greatly stabilizes the training process. The scalability and flexibility of DiT is empirically validated. DiT becomes the new backbone for diffusion models. In U-ViT, as shown in Figure 12, they treat all inputs, including time, condition, and noisy image patches, as tokens and propose long skip connections between the shallow and deep transformer layers. The results suggest that the downsampling and upsampling operators in CNN-based U-Net are not always necessary, and U-ViT achieves record-breaking FID scores in image and text-to-image generation.

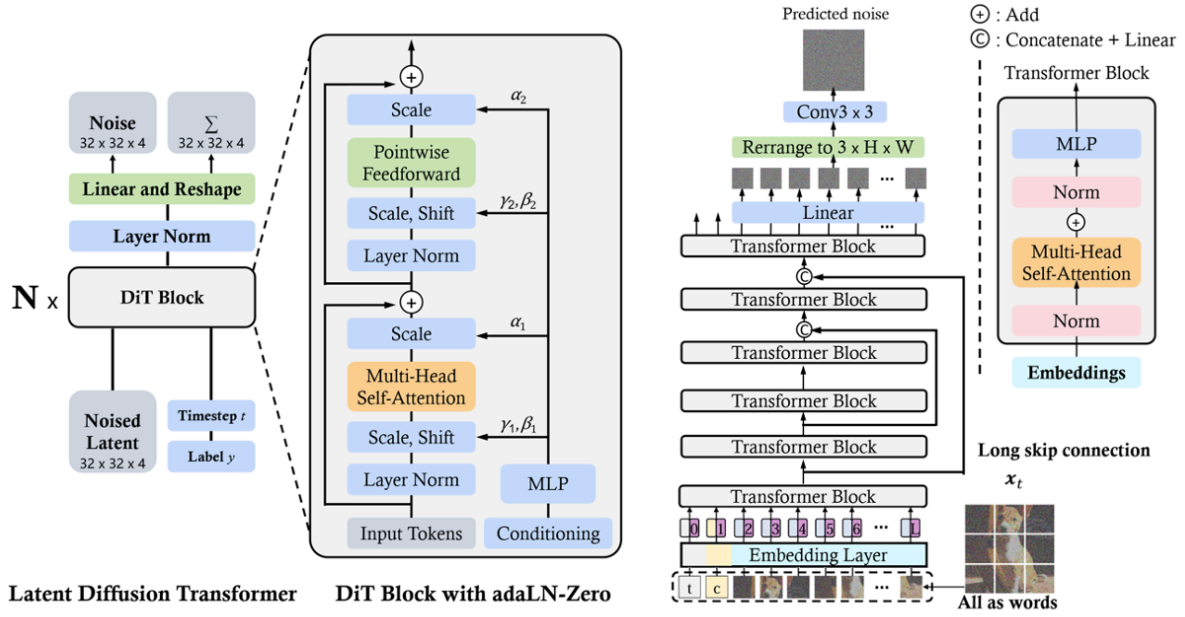


Figure 12: The overall framework of DiT (left) and U-ViT (right)

Like Masked AutoEncoder (MAE), Masked Diffusion Transformer (MDT) incorporates mask latent modeling into the diffusion process to explicitly enhance contextual relation learning among object semantic parts in image synthesis. Specifically, as shown in Figure 13, MDT uses a side-interpolated for an additional masked token prediction task during training to boost the training efficiency and learn powerful context-aware positional embedding for inference. Compared to DiT, MDT achieves better performance and faster learning speed. In MDTv2, they further improve MDT with a more efficient macro network with some u-shape long-shortcuts in the encoder and dense shortcuts from encoder input in the decoder. Furthermore, they also leverage a bag of training strategies like the Adan optimizer, Min-SNR weighting, dynamic masking ratio, and an improved power-cosine weighting for classifier-free guidance. MDTv2 achieves a significant boost in synthesis performance and learning speed. Instead of using AdaLN (i.e., shifting and scaling) for time-conditioning modeling, Hatamizadeh et al. introduce Diffusion Vision Transformers (DiffiT), which uses a time-dependent self-attention (TMSA) module to model dynamic denoising behavior over sampling time steps. Besides, DiffiT uses two hybrid hierarchical architectures for efficient denoising in the pixel space and the latent space, respectively, and achieves new state-of-the-art results across various generation tasks. Overall, these studies show promising results in employing vision transformers for image latent diffusion, paving the way for future studies for other modalities.

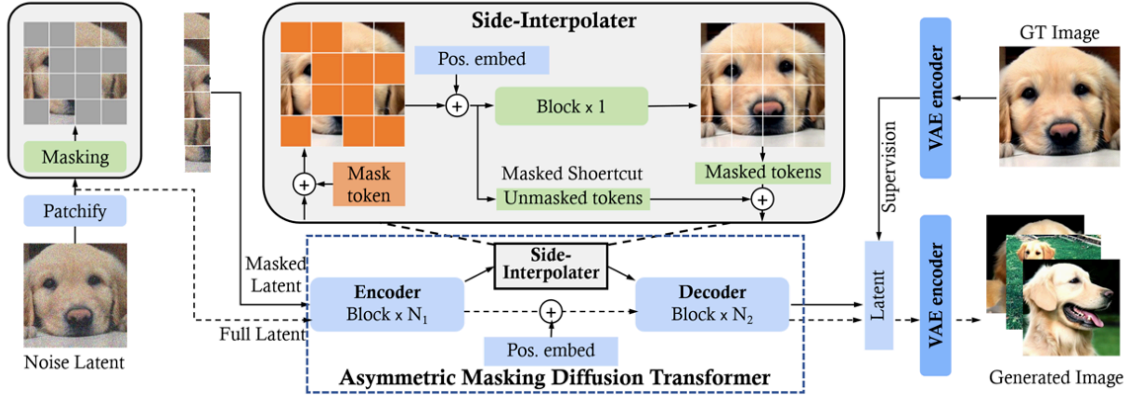


Figure 13: The overall framework of Masked Diffusion Transformer (MDT). A solid/dotted line indicates the training/inference process for each time step. Masking and side-interpolator are only used during training and are removed during inference.

Video Diffusion Transformer: Building upon the foundational works in text-to-image (T2I) diffusion models, recent research has been focused on realizing the potential of diffusion transformers for text-to-video (T2V) generation tasks. Due to the temporal nature of videos, key challenges for applying DiTs in the video domain are: *i) how to compress the video spatially and temporally to a latent space for efficient denoising; ii) how to convert the compressed latent to patches and feed them to the transformer; and iii) how to handle long-range temporal and spatial dependencies and ensure content consistency.* Please refer to Section 6.2.3 for the first challenge. In this section, we focus our discussion on transformer-based denoising network architectures designed to operate in the spatially and temporally compressed latent space. We give a detailed review of the two important works (Imagen Video and Video LDM) shown in the OpenAI Sora technique report and also briefly introduce some of recent promising approaches.

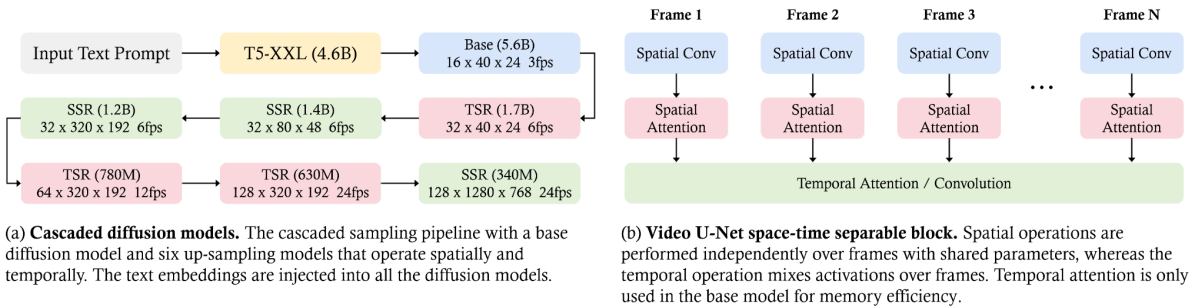


Figure 14: The overall framework of Imagen Video. Source: Imagen Video

Imagen Video, a text-to-video generation system developed by Google Research, utilizes a cascade of diffusion models, which consists of 7 sub-models that perform text-conditional video generation, spatial super-resolution, and temporal super-resolution, to transform textual prompts into high-definition videos. As shown in Figure 14, firstly, a frozen T5 text encoder generates contextual embeddings from the input text prompt. These embeddings are critical for aligning the generated video with the text prompt and are injected into all models in the cascade, in addition to the base model. Subsequently, the embedding is fed to the base model for low-resolution video generation, which is then refined by cascaded

diffusion models to increase the resolution. The base video and super-resolution models use a 3D U-Net architecture in a spatial temporal separable fashion. This architecture weaves temporal attention and convolution layers with spatial counterparts to efficiently capture inter-frame dependencies. It employs v-prediction parameterization for numerical stability and conditioning augmentation to facilitate parallel training across models. The process involves joint training on both images and videos, treating each image as a frame to leverage larger datasets, and using classifier-free guidance to enhance prompt fidelity. Progressive distillation is applied to streamline the sampling process, significantly reducing the computational load while maintaining perceptual quality. Combining these methods and techniques allows Imagen Video to generate videos with not only high fidelity but also remarkable controllability, as demonstrated by its ability to produce diverse videos, text animations, and content in various artistic styles.

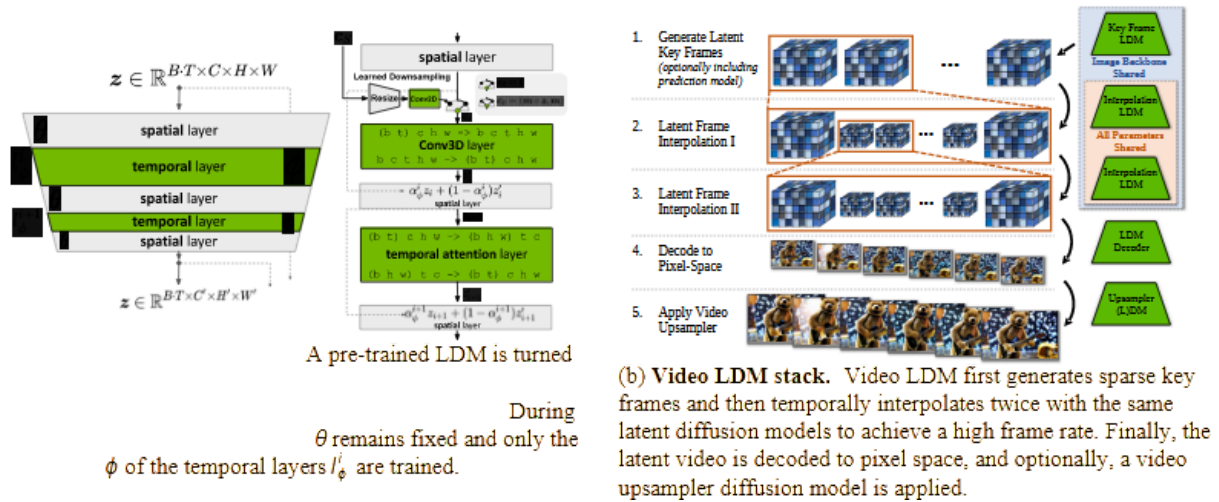


Figure 14: The overall framework of Video LDM. Source: Video LDM

Blattmann et al. [36] propose to turn a 2D Latent Diffusion Model into a Video Latent Diffusion Model (Video LDM). They achieve this by adding some post-hoc temporal layers among the existing spatial layers into both the U-Net backbone and the VAE decoder that learns to align individual frames. These temporal layers are trained on encoded video data, while the spatial layers remain fixed, allowing the model to leverage large image datasets for pre-training. The LDM's decoder is fine-tuned for temporal consistency in pixel space and temporally aligning diffusion model upsamplers for enhanced spatial resolution. To generate very long videos, models are trained to predict a future frame given a number of context frames, allowing for classifier-free guidance during sampling. To achieve high temporal resolution, the video synthesis process is divided into key frame generation and interpolation between these keyframes. Following cascaded LDMs, a DM is used to further scale up the Video LDM outputs by four times, ensuring high spatial resolution while maintaining temporal consistency. This approach enables the generation of globally coherent long videos in a computationally efficient manner. Additionally, the authors demonstrate the ability to transform pre-trained image LDMs (e.g., Stable Diffusion) into text-to-video models by training only the temporal alignment layers, achieving video synthesis with resolutions up to 1280×2048 px. Recently, there are also some other promising works that show superior performance in high-quality and temporal-consistent video generation. Similar to Imagen Video, LAVIE [62] proposes to transform a T2I model into a T2V models with cascaded LDMs, with two key insights: i) using simply temporal self-attention and RoPE [63] can adequately

capture temporal correlations inherent in video data; ii) joint image-video fine-tuning enable more stable training and avoid catastrophic forgetting. GenTron adapts pre-trained class-conditioning DiTs to textual conditioning and explores different conditioning mechanisms, where they find that cross attention is a better conditioning method compared to previous adaLM-Zero for free-form textual conditioning. Furthermore, by designing motion-free masking in the inserted temporal self-attention layers, GenTron is able to do efficient image-video joint training with control on the temporal modeling ability in the video diffusion process. Window Attention Latent Transformer (W.A.L.T) proposes to unify the latent space of both videos and images with 3D causal CNN and a special coding strategy for the first frame. Moreover, W.A.L.T uses non-overlapping, window-restricted spatial and spatiotemporal attention to lower the computational cost and provide flexibility for image-video joint training. Latent Diffusion Transformer (Latte) conducts extensive experiments among four model variants from the perspective of decomposing spatial and temporal information in video data. Furthermore, Latte studies the effect of a wide range of training strategies and modules in a DiT-based video generation system, including conditioning mechanisms and temporal positional embedding. Instead of using separate compression for spatial and temporal dimensions in previous works, SnapVideo proposes to extend the EDM framework for high-dimensional video data. Furthermore, SnapVideo treats images as T frames videos with infinite frame rates and introduces a variable frame-rate training procedure to bridge the modality gap. For scaling up to billion-parameters size, SnapVideo uses an efficient transformer-based architecture, FITs, as a scalable video generator, and a model cascade consisting of a first-stage lower-solution base model ($36 \times 64\text{px}$) and a high-solution upsampler ($288 \times 512\text{px}$) with corrupt training strategy.

6.4 Language Instruction Following

Users primarily engage with generative AI models through natural language instructions, known as text prompts. Model instruction tuning aims to enhance AI models' capability to follow prompts accurately. This improved capability in prompt following enables models to generate output that more closely resembles human responses to natural language queries. We start our discussion with a review of instruction following techniques for large language models (LLMs) and text-to-image models such as DALL·E 3. To enhance the text-to-video model's ability to follow text instructions, Sora utilizes an approach similar to that of DALL·E 3. The approach involves training a descriptive captioner and utilizing the captioner's generated data for fine-tuning. As a result of instruction tuning, Sora is able to accommodate a wide range of user requests, ensuring meticulous attention to the details in the instructions and generating videos that precisely meet users' needs.

6.4.1 Large Language Models

The capability of LLMs to follow instructions has been extensively explored. This ability allows LLMs to read, understand, and respond appropriately to instructions describing an unseen task without examples. Prompt following ability is obtained and enhanced by fine-tuning LLMs on a mixture of tasks formatted as instructions, known as instruction tuning. Wei et al. showed that instruction-tuned LLMs significantly outperform the untuned ones on unseen tasks. The instruction-following ability transforms LLMs into general-purpose task solvers, marking a paradigm shift in the history of AI development.

6.4.2 Text-to-Image

The instruction following in DALL·E 3 is addressed by a caption improvement method with a hypothesis that the quality of text-image pairs that the model is trained on determines the performance of the resultant text-to-image model. The poor quality of data, particularly the prevalence of noisy data and short captions that omit a large amount of visual information, leads to many issues such as neglecting keywords and word order, and misunderstanding the user intentions. The caption improvement approach addresses these issues by re-captioning existing images with detailed, descriptive captions. The approach first trains an image captioner, which is a vision-language model, to generate precise and descriptive image captions. The resulting descriptive image captions by the captioner are then used to fine-tune text-to-image models. Specifically, DALL·E 3 follows contrastive captioners (CoCa) to jointly train an image captioner with a CLIP architecture and a language model objective. This image captioner incorporates an image encoder, a unimodal text encoder for extracting language information, and a multimodal text decoder. It first employs a contrastive loss between unimodal image and text embeddings, followed by a captioning loss for the multimodal decoder's outputs. The resulting image captioner is further fine-tuned on a highly detailed description of images covering main objects, surroundings, backgrounds, texts, styles, and colorations. With this step, the image captioner is able to generate detailed descriptive captions for the images. The training dataset for the text-to-image model is a mixture of the re-captioned dataset generated by the image captioner and ground-truth human-written data to ensure that the model captures user inputs. This image caption improvement method introduces a potential issue: a mismatch between the actual user prompts and descriptive image descriptions from the training data. DALL·E 3 addresses this by upsampling, where LLMs are used to re-write short user prompts into detailed and lengthy instructions. This ensures that the model's text inputs received in inference time are consistent with those in model training.

6.4.3 Text-to-Video

To enhance the ability of instruction following, Sora adopts a similar caption improvement approach. This method is achieved by first training a video captioner capable of producing detailed descriptions for videos. Then, this video captioner is applied to all videos in the training data to generate high-quality (video, descriptive caption) pairs, which are used to fine-tune Sora to improve its instruction following ability. Sora's technical report does not reveal the details about how the video captioner is trained. Given that the video captioner is a video-to-text model, there are many approaches to building it. A straightforward approach is to utilize CoCa architecture for video captioning by taking multiple frames of a video and feeding each frame into the image encoder, known as VideoCoCa. VideoCoCa builds upon CoCa and re-uses the image encoder pre-trained weights and applies it independently on sampled video frames. The resulting frame token embeddings are flattened and concatenated into a long sequence of video representations. These flattened frame tokens are then processed by a generative pooler and a contrastive pooler, which are jointly trained with the contrastive loss and captioning loss. Other alternatives to building video captioners include mPLUG-2, GIT, FrozenBiLM, and more. Finally, to ensure that user prompts align with the format of those descriptive captions in training data, Sora performs an additional prompt extension step, where GPT-4V is used to expand user inputs to detailed descriptive prompts.

7. Examples



8. Opportunities

Academy: (1) The introduction of Sora by OpenAI marks a strategic shift towards encouraging the broader AI community to delve deeper into the exploration of text-to-video models, leveraging both diffusion and transformer technologies. This initiative aims to redirect the focus toward the potential of creating highly sophisticated and nuanced video content directly from textual descriptions, a frontier that promises to revolutionize content creation, storytelling, and information sharing.

(2) The innovative approach of training Sora on data at its native size, as opposed to the traditional methods of resizing or cropping, serves as a groundbreaking inspiration for the academic community. It opens up new pathways by highlighting the benefits of utilizing unmodified datasets, which leads to the creation of more advanced generative models.

Industry: (1) The current capabilities of Sora signal a promising path for the advancement of video simulation technologies, highlighting the potential to significantly enhance realism within both physical and digital areas. The prospect of Sora enabling the creation of highly realistic environments through textual descriptions presents a promising future for content creation. This potential extends to revolutionizing game development, offering a glimpse into a future where immersive-generated worlds can be crafted with unprecedented ease and accuracy.

(2) Companies may leverage Sora to produce advertising videos that swiftly adapt to market changes and create customized marketing content. This not only reduces production costs but also enhances the appeal and effectiveness of advertisements. The ability of Sora to generate highly realistic video content from textual descriptions alone could revolutionize how brands engage with their audience, allowing for the creation of immersive and compelling videos that capture the essence of their products or services in unprecedented ways.

Society: (1) While the prospect of utilizing text-to-video technology to replace traditional filmmaking remains distant, Sora and similar platforms hold transformative potential for content creation on social media. The constraints of current video lengths do not diminish the impact these tools can have in making high-quality video production accessible to everyone, enabling individuals to produce compelling content without the need for expensive equipment. It represents a significant shift towards empowering content creators across platforms like TikTok and Reels, bringing in a new age of creativity and engagement.

(2) Screenwriters and creative professionals can use Sora to transform written scripts into videos, assisting them in better showing and sharing their creative concepts, and even in producing short films and animations. The ability to create detailed, vivid videos from scripts can fundamentally change the pre-production process of filmmaking and animation, offering a glimpse into how future storytellers might pitch, develop, and refine their narratives. This technology opens up possibilities for a more dynamic and interactive form of script development, where ideas can be visualized and assessed in real time, providing a powerful tool for creativity and collaboration.

(3) Journalists and news organizations can also utilize Sora to quickly generate news reports or explanatory videos, making the news content more vivid and engaging. This can significantly increase the coverage and audience engagement of news reports. By providing

a tool that can simulate realistic environments and scenarios, Sora offers a powerful solution for visual storytelling, enabling journalists to convey complex stories through engaging videos that were previously difficult or expensive to produce. In summary, Sora's potential to revolutionize content creation across marketing, journalism, and entertainment is immense.

9. Conclusion